

Further Mechanics 1

Notes · Edexcel A Level Further Mathematics — Further Mechanics 1 · Chapter 1: Momentum and Impulse

Chapter 1: Momentum and Impulse

Why this chapter matters

Why this matters. Everything else in Further Mechanics 1 — springs, collisions, oblique impacts — is built on momentum and impulse. Get this chapter watertight (especially sign convention and vector direction) and the rest of the unit becomes an exercise in applying the same idea again and again.

Concept: Momentum

The momentum of a particle of mass m moving with velocity v is:

$$\mathbf{p} = m\mathbf{v}$$

Momentum is a **vector** — it has a direction as well as a magnitude, measured in kg m s^{-1} (equivalently N s). In one-dimensional problems, you must fix a positive direction at the start and stick to it: velocities (and hence momenta) in the opposite direction are negative.

Key vocabulary

Term	Meaning
Impulse (I or J)	The change in momentum caused by a force acting over a time interval, or by a collision
Direct collision	A collision in which both particles move along the same straight line, before and after impact (as opposed to Chapter 5's oblique, two-dimensional impacts)
Coalesce	Two particles collide and stick together, moving afterwards with a common velocity
Conservation of momentum	In a collision with no external horizontal force, total momentum before = total momentum after

Key formula: Impulse–momentum equation

Quantity	Formula & notes
Momentum	$\mathbf{p} = m\mathbf{v}$ — vector, kg m s^{-1}
Impulse of a constant force	$I = Ft$ — F constant, t = time acting

Impulse–momentum principle	$I = mv - mu$ — impulse = change in momentum
Conservation of momentum (2 particles)	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ — u = before, v = after

The impulse–momentum principle is really Newton's second law integrated over time: $F = ma = m(v - u)/t \Rightarrow Ft = mv - mu$. This is why impulse has the same units as momentum.

Physical intuition. Impulse is what a force “delivers” over a stretch of time — a small force acting for a long time can produce the same impulse (and the same change in momentum) as a large force acting briefly. This is why bending your knees when you land from a jump reduces the force on them: your momentum change is fixed by your mass and landing speed, so stretching that change over a longer time reduces the force needed to produce it. When a force varies with time rather than staying constant, impulse is the area under its force–time graph — a skill that reappears in Chapter 3, where tension itself varies as a spring stretches.

Concept: Conservation of momentum in direct collisions

When two particles collide and no external horizontal force acts (friction, gravity along the line of motion, etc. are ignored unless stated), the **total momentum of the system is unchanged** by the collision:

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

This is typically one equation in one unknown, once you're told what happens (e.g. they coalesce, or one velocity after is given). For a *pair* of unknowns after a collision, you need a second equation — this is exactly what Newton's law of restitution supplies in Chapter 4. Note that conservation of momentum says nothing about kinetic energy — unless a collision is perfectly elastic, some KE is always lost (into heat, sound and deformation). Chapter 4 introduces the coefficient of restitution, e , which controls exactly how much.

Worked example 1 — direct collision, particles coalesce

EXAMPLE 1.1

Question. A particle A of mass 3 kg moving at 8 m s^{-1} collides directly with a particle B of mass 5 kg moving at 2 m s^{-1} in the same direction. On collision, A and B coalesce. Find the common velocity after the collision, and the impulse exerted by A on B.

Step 1. Define positive direction as the direction of motion of A. Momenta before: A has $3 \times 8 = 24 \text{ kg m s}^{-1}$, B has $5 \times 2 = 10 \text{ kg m s}^{-1}$. **B1**

Step 2. Apply conservation of momentum, letting v be the common velocity after coalescing (combined mass 8 kg):

$$3(8) + 5(2) = 8v \quad \text{M1}$$

$$24 + 10 = 8v \Rightarrow v = 34/8 = 4.25 \text{ m s}^{-1} \quad \text{A1}$$

Step 3. Impulse exerted by A on B equals the change in momentum of B:

$$J = m_B(v - u_B) = 5(4.25 - 2) = 5 \times 2.25 = 11.25 \text{ N s} \quad \text{M1A1}$$

(By Newton's third law, the impulse exerted by B on A is -11.25 N s — equal and opposite.)

Answer: common velocity 4.25 m s^{-1} in A's original direction; impulse on B from A = 11.25 N s .

EXAM TIP

Marks are very often lost by forgetting that momentum is a vector. Before any collision problem, get into the habit of drawing a single arrow labelled “+ direction” on your diagram. Any velocity given “in the opposite direction” must be substituted as a *negative* number — do this substitution explicitly in the working, don't let it happen silently in your head, or errors creep in on the next line.

Worked example 2 — direct collision, particles moving towards each other, separate afterwards

EXAMPLE 1.2

Question. A particle P of mass 2 kg moving at 6 m s^{-1} collides directly with a particle Q of mass 4 kg moving at 3 m s^{-1} in the opposite direction. After the collision, Q moves at 1 m s^{-1} in the direction it was originally travelling. Find the velocity of P after the collision, stating clearly its direction, and find the magnitude of the impulse on P.

Step 1. Take the direction of P's initial motion as positive. Then $u_P = 6$, $u_Q = -3$ (opposite direction). **B1**

Step 2. After the collision, Q still travels in its original direction, so $v_Q = -1$. Apply conservation of momentum:

$$2(6) + 4(-3) = 2v_P + 4(-1) \quad \text{M1}$$

$$12 - 12 = 2v_P - 4$$

$$0 = 2v_P - 4 \implies v_P = 2 \text{ m s}^{-1} \quad \text{A1}$$

Step 3. Since $v_P = +2$, P now moves at 2 m s^{-1} in its *original* direction (it has slowed down but not reversed). **A1**

Step 4. Impulse on P: $J = m_P(v_P - u_P) = 2(2 - 6) = -8 \text{ N s}$. Magnitude 8 N s , direction opposite to P's original motion. **M1A1**

Answer: P moves at 2 m s^{-1} in its original direction after impact; the impulse on P has magnitude 8 N s , directed opposite to P's initial velocity.

EXAM TIP

This example is worth doing early because it forces you to handle a genuinely negative given velocity (Q's) as well as a possibly negative *found* velocity (P's, if the collision had reversed it). Before calculating, predict the sign of v_P and justify it physically — it builds the habit of sanity-checking an algebraic answer against the physical picture, which examiners reward and which catches sign slips immediately.

Chapter summary

Term	Meaning
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Momentum	$p = mv$, a vector; direction matters and must be fixed at the outset
Impulse of constant force	$I = Ft$
Impulse–momentum principle	$I = mv - mu$, links force/time to change in velocity
Conservation of momentum	$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$, valid when no external horizontal force acts
Newton's third law link	Impulse of A on B = $-(\text{impulse of B on A})$

Common exam mistakes

- Forgetting to assign a positive direction before starting, then mixing signs inconsistently partway through.
- Treating “coalesce” as meaning the velocities average rather than using conservation of momentum with the combined mass.
- Reporting only the magnitude of a final velocity when the question explicitly asks for direction too.
- Confusing impulse on A with impulse on B — they are equal in magnitude but opposite in sign (Newton's third law), and examiners frequently ask for one specific one.
- Quoting units for impulse or momentum incorrectly (it is N s or kg m s^{-1} , not N or kg m s^{-2}).

You should now be able to

- State and use $p = mv$ and the impulse–momentum equation $I = mv - mu$.
- Apply conservation of momentum to a direct collision between two particles, including coalescence.
- Correctly assign and maintain a sign convention throughout a one-dimensional collision problem.
- Calculate the impulse exerted by one particle on another during a collision, and relate it to Newton's third law.